

BOOK A DOCTOR USING MERN

Full Stack Development Documentation

1. Introduction

- Project Title: Book a Doctor Using MERN
- Domain: Healthcare Management System
- Technology: MongoDB, Express.js, React.js, Node.js
- Objective: To provide an online platform for patients to search doctors, book appointments, upload medical records, and manage consultations digitally.

2. Project Overview

Purpose

The system simplifies healthcare appointment management by connecting patients, doctors, and administrators through a centralized web platform.

Key Features

- User Registration & Login
- Doctor Registration & Approval
- Doctor Search & Filtering
- Appointment Booking
- Appointment History
- Notifications
- Document Upload
- Admin Dashboard
- Doctor Dashboard

3. System Requirements

Software Requirements

- Windows 10/11, Linux, macOS
- Node.js v16+
- npm v8+
- React.js
- Express.js
- MongoDB
- VS Code
- Git & GitHub
- Postman

Hardware Requirements

- Intel i5 / Ryzen 5
- 8 GB RAM minimum
- 1 GB Storage
- 1366×768 Display

4. Technology Stack

Layer	Technology
Frontend	React.js, Bootstrap, Material UI
Backend	Node.js, Express.js
Database	MongoDB
Authentication	JWT, bcrypt
API Client	Axios
Date Handling	Moment.js

5. Project Architecture

Frontend Layer

- React Components
- Axios API Calls
- Material UI

Backend Layer

- Express REST APIs
- Controllers
- Middleware

Database Layer

- MongoDB Collections
- Mongoose Models

6. MVC Architecture

Model Layer

- User Model
- Doctor Model
- Appointment Model

View Layer

- React Frontend
- User Dashboard
- Doctor Dashboard

- Admin Dashboard

Controller Layer

- User Controller
- Doctor Controller
- Admin Controller

7. Database Design

Users Collection

- user_id
- name
- email
- password
- phone
- notifications
- role

Doctors Collection

- doctor_id
- user_id
- specialization
- fees
- experience
- timings
- status

Appointments Collection

- appointment_id
- doctorInfo
- userInfo
- date
- status
- document

8. ER Diagram

(Insert ER diagram from original PDF)

9. User Roles

Patient

- Register/Login
- Search Doctors

- Book Appointments
- Upload Documents
- View History

Doctor

- Manage Profile
- View Appointments
- Approve/Reject Requests
- Update Status

Admin

- Approve Doctors
- Manage Users
- Monitor Appointments

10. Functional Requirements

Patient Module

- Registration
- Authentication
- Doctor Search
- Appointment Booking

Doctor Module

- Application Submission
- Appointment Management
- Profile Updates

Admin Module

- Doctor Approval
- User Monitoring
- Appointment Monitoring

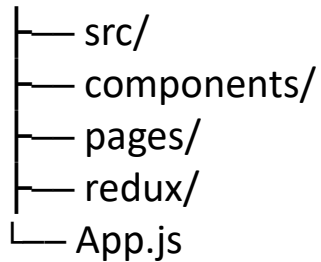
11. User Flow

1. User Registration
2. Login
3. Doctor Search
4. Appointment Booking
5. Doctor Approval
6. Appointment Confirmation
7. Consultation
8. Follow-up

12. Folder Structure

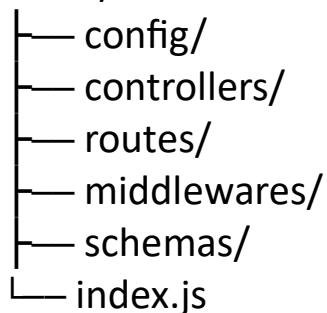
Client

client/



Server

server/



13. Backend Design

index.js

- Express Initialization
- Middleware Setup
- Route Registration
- Server Startup

connectToDB.js

- MongoDB Connection
- Error Handling

Controllers

- User Controller
- Doctor Controller
- Admin Controller

14. API Documentation

User APIs

- POST /register
- POST /login
- GET /profile

Doctor APIs

- POST /apply-doctor
- GET /appointments
- PUT /status-update

Admin APIs

- GET /users
- GET /doctors
- PUT /approve-doctor
- PUT /reject-doctor

15. Authentication & Security

- JWT Authentication
- Password Hashing using bcrypt
- Protected Routes
- Role-Based Access Control

16. Appointment Management Module

- Appointment Creation
- Status Updates
- History Tracking
- Notification Management

17. Notification System

- Appointment Approved
- Appointment Rejected
- Appointment Scheduled
- Appointment Completed

18. Testing

- API Testing using Postman
- Functional Testing
- UI Testing
- Authentication Testing

19. Security Features

- JWT Tokens
- Password Encryption
- Protected APIs
- Access Control

20. Performance & Scalability

- MongoDB Scalability

- Load Balancing
- Caching
- Optimized Queries

21. Known Issues

- Limited Real-Time Communication
- No Video Consultation
- No Payment Gateway

22. Future Enhancements

- AI Health Recommendations
- Online Payments
- Video Consultation
- Mobile Application
- SMS Integration
- Electronic Health Records

23. Conclusion

The Book a Doctor platform provides a complete healthcare appointment management system using the MERN stack. It improves accessibility, reduces manual effort, and creates an efficient interaction platform between patients, doctors, and administrators.